The U-Shaped Challenge of Language Grounding: How Instruction Granularity Shapes Instruction Following Agent Behavior

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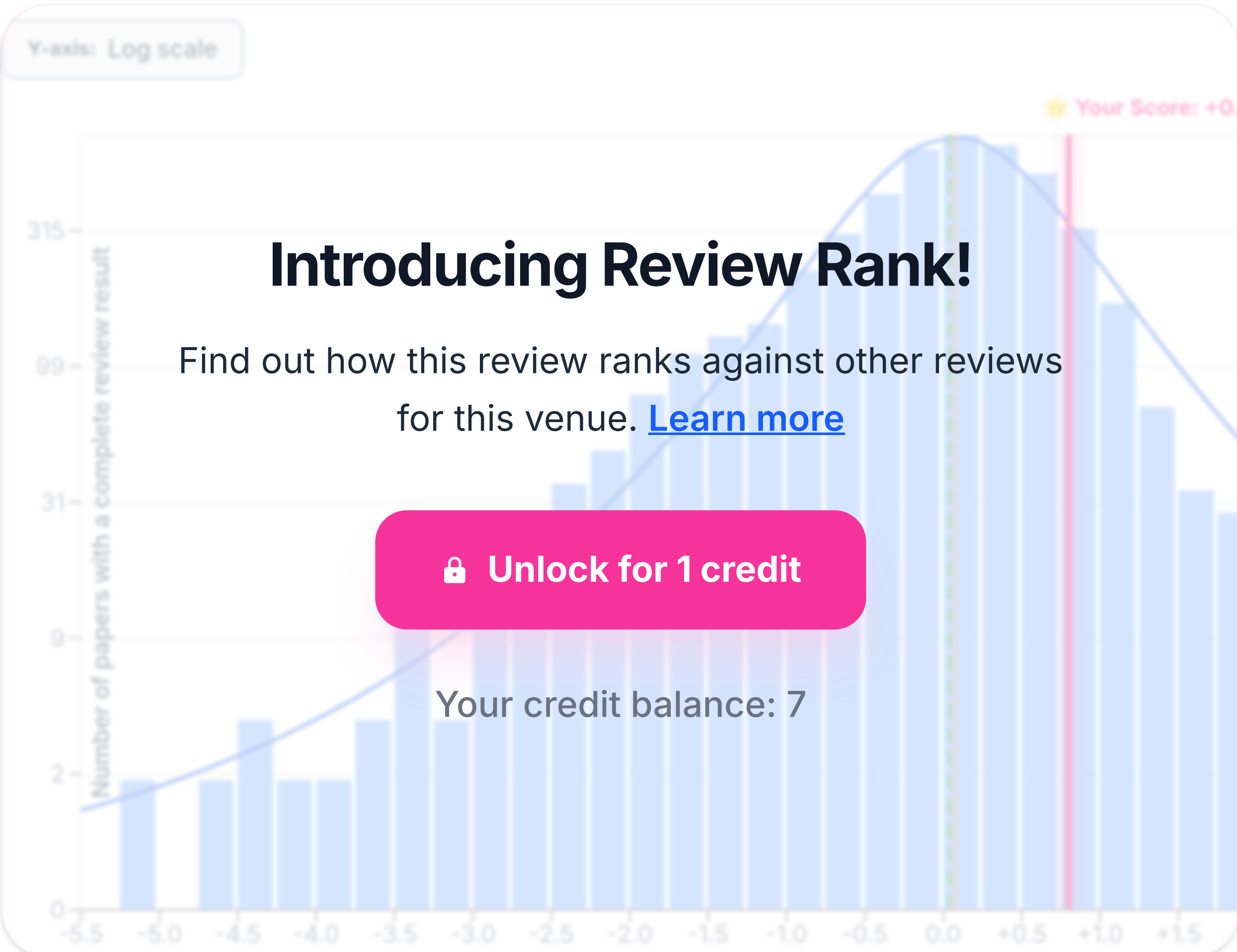
Reviewed as if submitted to CONFERENCE ACL 2026 - main track



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Desk Rejection Assessment:

Paper Length

Pass ✅. The paper is within the standard length limits for an ACL main track submission.

Topic Compatibility

Pass ✅. The paper's focus on language grounding, instruction following, and embodied AI aligns perfectly with the ACL topic "Multimodality and Language Grounding to Vision, Robotics and Beyond."

Minimum Quality

Pass ✅. The paper is well-structured with all essential sections (Abstract, Introduction, Related Work, Methodology, Experiments, Results, Conclusion, Limitations). It is written in clear academic English, and its methodology, while having limitations, appears internally consistent and sufficiently detailed for a rigorous review. The work presents a novel research question with empirical validation, meeting the minimum quality standards for ACL.

Prompt Injection and Hidden Manipulation Detection

Pass ✅. A thorough check of the paper's content reveals no hidden prompts, invisible text, or any other attempts to manipulate the review process.

Expected Review Outcome:

Paper Summary

This paper proposes a formal, agent-centric metric for instruction granularity called "rule width" (w), adapted from width-based planning. The authors introduce a new benchmark, Mini-Behavior-Gran, to systematically study the impact of w on vision-language-action (VLA) models in a controlled embodied AI environment. Their key finding is a U-shaped performance curve: agents perform well on very fine-grained ($w\leq 1$) and coarse-grained ($w\geq 2$) instructions, but struggle in the middle. The paper argues this is due to a shift from deep semantic grounding at low width to shallow, vision-based shortcuts at high width, driven by instruction predictability. The authors also report an asymmetric generalization ability and the benefits of mixed-granularity training.

Summary Of Strengths

This paper presents a compelling and well-executed study on a fundamental, yet under-formalized, aspect of language-guided embodied AI.

The primary strength is the introduction of "rule width" (w) as a principled, cross-task metric for instruction granularity. The field has long relied on vague, text-centric heuristics like sentence length. By connecting granularity to the agent's internal decision-making burden (i.e., the number of concurrent state features to track), the paper provides a much-needed formal tool for analysis. The motivation in Sections 1 and 3 is clear and convincingly argues for moving beyond surface-level metrics. The running example in Table 1 and Section 3.1 effectively illustrates the concept.

A second major strength is the creation of the Mini-BEHAVIOR-Gran benchmark. Designing a controlled environment to isolate a single variable (granularity) while keeping others constant is commendable and enables the kind of rigorous scientific inquiry often missing in this domain. The experimental pipeline, detailed in Figure 2, is well-designed to ensure that observed performance differences are attributable to grounding ability rather than confounding factors.

The empirical findings are insightful and, in some cases, counter-intuitive. The discovery of the non-monotonic, U-shaped performance curve (Figure 5) is the paper's most significant result. It challenges the simple assumption that coarser instructions are always harder. The authors don't just report this finding; they provide a well-reasoned and empirically supported mechanistic explanation: a trade-off between the cognitive load of grounding (measured by w) and the statistical predictability of the instruction ($P(I|v)$). This analysis, supported by the perturbation experiments in Figure 6 and the predictability measurements in Figure 7, is a strong piece of scientific detective work.

Finally, the paper reports other valuable findings, such as the asymmetric generalization from coarse-to-fine instructions (Table 2) and the identification of a hard "capacity cap" at $w \geq 4$. These results provide practical insights for training more robust instruction-following agents, particularly the demonstrated effectiveness of mixed-granularity training.

Summary Of Weaknesses

Despite its strengths, the paper has several significant weaknesses that temper its conclusions and raise questions about the generalizability and broader impact of the findings.

1. **Over-reliance on a Simplified, "Toy" Environment:** The entire empirical foundation of this paper rests on Mini-BEHAVIOR, a 2D grid-world environment. While the authors justify this choice for control and isolation (Appendix B), it is a major limitation. Real-world embodied AI involves grappling with partial observability, noisy perception, complex 3D physics, and continuous control spaces. By abstracting all of this away, it's unclear if the observed U-shaped phenomenon and the proposed mechanisms (e.g., visuo-motor shortcuts based on predictable grid patterns) would hold in more realistic settings. The "visuo-motor shortcuts" here are likely very different

from those in photorealistic simulators or the real world. This severely limits the external validity of the claims.

2. **The Oracle Instructor Undermines Claims about Planning:** The experimental setup (Figure 2) uses an "Oracle Instructor" that provides the agent with the *correct* rule to execute at each step. The authors argue this isolates the grounding challenge. However, it also fundamentally changes the problem. The agent is no longer performing hierarchical planning or high-level reasoning; it is reduced to a reactive policy that grounds single-step commands. This setup significantly weakens the claim that w measures "latent planning demand" for the *neural agent*. The planning has already been done by the oracle. The agent is simply executing a pre-sequenced sub-task, and the difficulty it faces is in low-level state coordination, not in deciding the sequence of operations.
3. **Ambiguous Definition and Measurement of "Shallow Grounding":** The paper diagnoses the performance rebound at high width as a shift to "shallow grounding" and "visuo-motor shortcuts." The evidence provided, primarily the $\Delta \mathrm{SR}$ from language perturbation experiments in Figure 6, is suggestive but not conclusive. The claim that the Pure-C policy's small performance drop (a 2.8% difference between "Static Goal Lang" and "Without Language") indicates shallow grounding is an interpretation. An alternative interpretation is that coarse instructions, by their nature, provide less information, so a policy trained on them naturally relies more on visual context to solve the task. The paper doesn't provide a direct, independent measure of "grounding depth" and instead infers it from task performance, which is somewhat circular. The argument would be much stronger with more direct probes, such as analyzing the model's internal representations or attention patterns.
4. **Overstatement of `w` as a Fundamental Bottleneck for Neural Agents:** The paper claims that the performance collapse at $w \geq 4$ (Table 2) suggests that `w` is a "fundamental representational bottleneck shared by both symbolic and neural paradigms." This is a strong claim based on correlational evidence from one specific, simplified domain and a particular set of models. The collapse could be an artifact of the dataset, the specific VLM backbone used, or the training regime. To claim this is a fundamental property of neural agents in general is an overgeneralization.
5. **Limited Exploration of VLA Architectures:** The paper evaluates three "action-decoding archetypes" (AR, Diffusion, Parallel), which is a good step. However, these are all variations on a similar theme within a single VLM backbone (Prismatic-7B). The findings would be more robust if tested across different families of models (e.g., smaller, non-LLM based agents, or agents with explicit memory modules) to confirm if the U-shaped trend is a universal phenomenon or specific to this class of large VLMs.

6. **Incomplete Literature Review:** The paper fails to cite and discuss a substantial amount of directly relevant work on instruction following, language grounding, and the impact of instruction properties on agent behavior. This oversight weakens the paper's positioning and suggests the authors may not be fully aware of the existing landscape. The missing references are detailed in the "Potentially Missing Related Work" section. This is a significant weakness for a paper submitted to a top-tier venue.
7. **Clarity of the Running Example Calculation:** While the concept of width is explained, the calculation in Table 1 for the running example is dense and could be clearer. For instance, the transition from width 3 to 4 when cleaning the second shoe ("one has to track the same features as before, plus maintaining the cleanliness of the first shoe (C_1)") is the crux of the complexity increase. This could be explained more explicitly, as it's central to how width scales with task structure.

Potentially Missing Related Work

The paper's literature review is missing several highly relevant works that should be discussed to properly contextualize the contribution.

- 1. Pugacheva, D., et al., 'Bring the Apple, Not the Sofa: Impact of Irrelevant Context in Embodied AI Commands on VLA Models', 2025 — This paper directly investigates how variations in command content (specifically, irrelevant context) affect VLA models, which is closely related to the theme of instruction properties influencing agent behavior. It should be discussed in Section 2 when covering language sensitivity.
- 2. Li, D., et al., 'Learning Instruction-Guided Manipulation Affordance via Large Models for Embodied Robotic Tasks', 2024 — This work explores how language instructions guide affordance prediction, a key aspect of grounding. It's relevant to the discussion of how instructions modulate the decision-making process and should be cited in Section 1 or 2.
- 3. Gao, J., et al., 'Compressor-VLA: Instruction-Guided Visual Token Compression for Efficient Robotic Manipulation', 2025 — This work deals with processing visual information conditioned on language instructions, touching on the interplay between language and perception that is central to the current paper's "shallow grounding" hypothesis. It could be mentioned in Section 2.
- 4. Nguyen, V.-Q., et al., 'Look Wide and Interpret Twice: Improving Performance on Interactive Instruction-following Tasks', 2021 — This paper's two-stage interpretation method is a form of decomposing the grounding problem, similar in spirit to how fine-grained instructions decompose tasks. It provides a relevant architectural perspective that should be acknowledged in Section 2.

- 5. Grazioso, M., & Suglia, A., 'An Analysis of Visually Grounded Instructions in Embodied AI Tasks', 2023 — This is a direct analysis of visually grounded instructions and would be a critical citation for situating the current work's contribution within the broader analysis of instruction following.
- 6. Körner, A., et al., 'Embodied Processing at Six Linguistic Granularity Levels: A Consensus Paper', 2023 — This consensus paper provides a foundational review of how language processing is affected by granularity, linking it to sensorimotor experience. Citing this would strengthen the theoretical underpinnings of Section 1 and 2.
- 7. Han, T., & Schlangen, D., 'Grounding Language by Continuous Observation of Instruction Following', 2017 — This earlier work on grounding from observation is relevant to the core problem and should be cited in the related work section to provide historical context.
- 8. Fried, D., 'Learning Grounded Pragmatic Communication', 2021 — This dissertation covers pragmatic inference in grounded domains, which relates to how agents should interpret underspecified (i.e., coarse) instructions. It's relevant to the theoretical framing in Section 1 and 2.
- 9. Huang, W., et al., 'Inner Monologue: Embodied Reasoning through Planning with Language Models', 2022 — This paper's use of language for explicit planning steps is a different approach to hierarchical decomposition than what is studied here, providing a useful contrast. It should be discussed in Section 2 alongside other hierarchical methods.
- 10. Goel, D., et al., 'Language Guided Meta-Control for Embodied Instruction Following', 2022 — This work proposes a meta-controller to improve language grounding, directly addressing the challenges of instruction following that this paper studies. It's a key related work that should be compared against.

Comments Suggestions And Typos

Here are some suggestions to improve the paper. My overall score could increase if the authors can convincingly address points 1 and 2 in their rebuttal.

1. **Addressing the "Oracle" Limitation:** Could you run an additional experiment where the agent must select the next rule from the set of available rules, instead of being given it by an oracle? This would introduce a hierarchical reasoning component and test whether the `w` of the *chosen* rule still correlates with failure. This would substantially strengthen your claims about "planning demand."
2. **Strengthening the "Shallow Grounding" Claim:** To make the claim less circular, could you provide more direct evidence? For example, analyzing the attention maps of the VLM to see if the model actually looks at the relevant objects less in the Pure-C setting. Or, could you use representation probes

(e.g., CKA) to show that the text and vision representations are less aligned in the coarse-trained models?

3. **Generalizability:** While running experiments in a more complex environment is likely out of scope for a rebuttal, it would be valuable to discuss the limitations of the grid-world more explicitly in the main paper (not just the appendix) and speculate on how the U-shaped trend might change in a 3D, physics-enabled environment. For instance, would the "rebound" at the coarse end disappear because reliable visuo-motor shortcuts are harder to learn?
4. **Typos/Clarity:**
   - Page 3, Table 1 Caption: "Cleaning k Shoes (k = 2)" is good. The formula "Width scales as w = k + 2" is very helpful.
   - Page 4, Section 3.4: "we assume the execution of r_i does not negate established fluents unless explicitly required by E_{r_i}". This is a key assumption (related to the frame problem) and is well-stated.
   - Page 6, Figure 4: The 'nan%' values are a bit jarring. It might be better to explain in the caption that these combinations did not occur in the dataset or resulted in zero successes.
   - The citations seem to use a mix of styles (e.g., "(Arjun et al., 2025)" vs " *(Dong and Lapata, 2018; Huang et al., 2025b)*"). Please ensure consistency.
   - Page 7, Figure 7: The caption says "VLM instruction-prediction Cross Entropy (CE) loss", but the y-axis is "Cross Entropy Loss". The title in the plot is "Cross Entropy Loss". The legend has "Coarse", "Medium", "Fine-grained". There's a slight inconsistency in terminology ("Fine" vs. "Fine-grained"). The box inside the plot says "Variance (Loss Range)" but then provides what looks like a single number per granularity, which is confusing. Is it the variance of the loss over the last few epochs, or the range? Please clarify.

My score would increase if the authors provide convincing arguments or preliminary results for suggestions 1 and 2, which would address the most critical weaknesses regarding the experimental design and the interpretation of the results.

Confidence

4 Quite sure after careful checking: I tried to check the important points carefully. It’s unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness

3.5 Between Acceptable and Strong: The study is methodologically sound within its self-imposed, controlled environment. The core claims are supported by the experiments presented. However, the external validity is limited by the simplified domain and the oracle setup, which makes the soundness of generalizing the conclusions questionable.

Excitement

4.0 Exciting: I would mention this paper to others and/or make an effort to attend its presentation in a conference. The formalization of granularity and the discovery of the U-shaped trend are novel and thought-provoking. The paper tackles a fundamental problem with a rigorous approach, and the findings challenge common intuitions.

Overall Assessment 📊 See how you rank ↑

3.5 Borderline Conference: The paper is close to being acceptable for the main conference but has notable weaknesses. The paper introduces a novel and valuable formalism (w) for instruction granularity and presents an intriguing set of findings, most notably the U-shaped performance curve. The analysis is thorough and insightful. However, the contribution is significantly constrained by the reliance on a simplified grid-world and an oracle instructor, which raises serious questions about the generalizability of the findings to more realistic embodied AI scenarios. The large number of missing citations also indicates a weakness in situating the work. While the ideas are strong, the empirical foundation needs to be more robust for an unconditional conference acceptance.

Best Paper Justification

N/A

Limitations And Societal Impact

The authors have a dedicated Limitations section (Section 7), which is good. They correctly identify the use of discrete action spaces as a limitation and provide a defense for the oracle instructor setup. However, the defense of the oracle is a bit weak; it claims the grounding gap is still present, but it sidesteps the fact that the agent's planning/sequencing abilities are not being tested. The discussion of limitations could be strengthened by more deeply engaging with the potential gap between the grid-world findings and what might happen in photorealistic 3D environments with continuous control. The paper does not discuss societal impact, which is acceptable given its foundational nature. There are no obvious negative societal impacts, as the work is focused on the mechanics of instruction following in a simulated context.

Ethical concerns

None. The work is conducted in a simulated environment with no human data, personally identifiable information, or risk of harm to individuals or communities. The authors have submitted the ARR Responsible NLP Checklist, and no ethical issues are apparent from the paper's content.

Needs Ethics Review

No.

Reproducibility

4 Mostly reproducible: They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method. The authors state they will release their benchmark (Mini-BEHAVIOR-Gran) and provide significant detail on their experimental setup, models, and hyperparameters in the appendix. The use of a controlled, symbolic environment should make replication more straightforward than in more complex simulators.

Datasets

4 Useful: I would recommend the new datasets to other researchers or developers for their ongoing work. The proposed Mini-BEHAVIOR-Gran benchmark, with its systematic variation of instruction width, would be a valuable resource for the community to study language grounding in a controlled manner. It enables targeted experiments that are difficult to conduct in existing, more complex benchmarks.

Software

3 Potentially useful.: Someone might find the new software useful for their work. The authors do not explicitly promise to release all their training and evaluation code, but if they release the code used to generate the benchmark and run the experiments, it would be useful for researchers wanting to build upon this work or test their own models in this controlled setting.

Reviewed on March 11, 2026 Gemini-2.5-Pro

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